

Choice of battery

Currently, two types of batteries are available for solar streetlighting.

Lead-acid batteries are the conventional options in the market. These batteries require constant maintenance and cannot be drained out completely while in use. These allow a discharge of up to 80 per cent. However, while lead-acid batteries require regular manual maintenance, these tend to have a longer life if maintained properly.

Lithium-ion and lithium-ferro phosphate batteries are the latest on the market. These batteries are much lighter (one-third that of lead-acid batteries), allow complete discharge during use and require almost no maintenance. However, their life is limited to nearly 3.5 years. Also, they cost 20-25 per cent more.

Dr Uppal suggests, "Lead-acid batteries are heavy and large in size, requiring them to be installed on the pole externally. These also need regular maintenance. So users need to install them at an accessible height on the pole, which makes them vulnerable to theft. Lithium-ion batteries have a high energy density and are maintenance-free."

What to buy?

Many buyers opt for AC LED streetlights owing to their lower investment cost and quicker ROI.

Dr Uppal shares, "Solar-based systems are ideal for Greenfields projects and remote locations where cabling cost savings, power savings and uninterrupted power (independent of grid) can provide a comparatively quicker ROI within 6 to 24 months, depending on the location."

Sambid Mohanty, AGM-Solar, Havells Lighting, explains benefits of solar streetlighting with an example: "If you notice AC-driven LED lightings across Noida-Greater Noida roads, a good number of them malfunction, the reason being poor maintenance and frequent power cuts. In areas like these, where cabling and grid power maintenance might be a challenge, investing in solar streetlights would be a better option." **EFY**

—Paromik Chakraborty, technical journalist, EFY

Mission 2030: Where Does India's EV Ecosystem Stand?

➤ **At present, India doesn't have any manufacturer with the technology or infrastructure to make lithium batteries for EVs**

India has started its journey towards achieving an all-green vehicle ecosystem by 2030. Transport minister Nitin Gadkari's bold statement of bulldozing in a completely green automotive era confirmed the priority of the agenda. English dailies quoted Gadkari as saying, "The government will be ready with its electric vehicle policy by December (2017) this year." Amidst all the rumble, however, is India prepared to face the shortcomings in its ecosystem as it inches towards success?

Strengths

India is going strong in many respects with regards to Mission 2030. Many initiatives have been taken by the administration to complement these strengths. For instance, Energy Effi-

ciency Services Limited (EESL) has incentivised ₹ 11.2 billion to Tata Motors and Mahindra and Mahindra Limited for production of 10,000 electric cars. Additionally, EESL is driving bids for setting up 4000 EV charging stations in Delhi NCR.

Over time, the increasing number of EVs and charging stations on the road would bring down their cost. Samrath S. Kochar, director, Trontek, affirms, "With bulk orders, the costs will come down significantly, making the vehicles commercially affordable. This will greatly magnify the adoption rate within India's population."

Take, for instance, Nagpur, which is the first Indian city to launch 200 EVs in 2017 including four-wheelers, buses and e-rickshaws. The city saw installation of over 50 charging stations, set up by Ola. The government has fixed Goods & Services Tax (GST) on automobiles accordingly—setting only 12 per cent GST on electric vehicles as against 28 per cent taxation on conventional petrol- and diesel-driven automobiles.

India's automobile components vertical can contribute well to the mission. Biju Bruno, MD, Greenvision Technologies, shares, "We have a strong components and sub-system manufacturing ecosystem for the automotive industry, which can adapt to making the necessary components for EVs."

Cummins India, an automobile engine manufacturer, has already started R&D on electric vehicle engines. Major auto-players SUN Mobility and Ashok Leyland are pairing up for electric-driven public transportation and private vehicle solutions.

Opportunities abound for energy solution companies as well. Setting up of charging infrastructure and manufacturing of efficient EV battery solutions, in particular, assume importance. Companies like Bharat Heavy Electricals Limited (BHEL) and National Thermal Power Corporation (NTPC) are working actively, with major support from the Indian Space Research Organisation (ISRO), to bring down the



EV vehicles hooked to charging stations